

Active Directory Lab – Part 1: Windows Server Installation and Initial Configuration

Objective

The objective of this lab was to install and configure a Windows Server virtual machine in preparation for an Active Directory domain environment. This included setting up the server in VirtualBox, confirming network connectivity, configuring a host-only lab network, and assigning a static IP address for future domain controller use.

Lab Environment

The lab was built using VirtualBox with a Windows Server virtual machine. The server was configured with two network adapters: one NAT adapter for basic internet connectivity and one host-only adapter for communication with other lab virtual machines.

The server was later prepared to act as the domain controller for the adlab.local domain.

Tools Used:

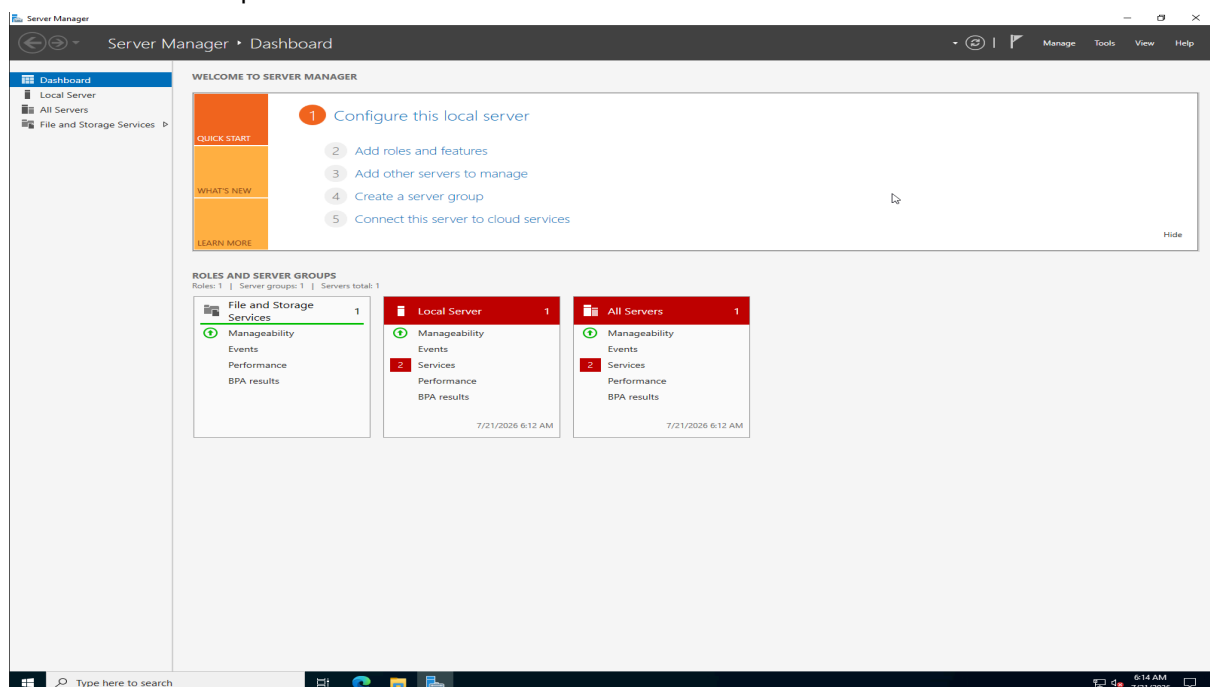
- Virtualbox
- Windows Server 2022 Evaluation
- Server Manager
- Command Prompt
- Windows Network Connections

Steps Performed

A new Windows Server 2022 virtual machine was created in VirtualBox. The virtual machine was named AD-DC-01 and configured with appropriate hardware resources for a small Active Directory lab, including 4 GB of memory, 2 CPU cores, and a virtual hard disk.

Windows Server 2022 Standard Evaluation with Desktop Experience was installed. Desktop Experience was selected so that the server could be managed using the graphical Windows interface and Server Manager.

After installation, Server Manager opened successfully, confirming that the Windows Server installation had completed.



The server was configured with two network adapters. The first adapter used NAT and received an IP address in the 10.0.2.0/24 range. This adapter was used for general connectivity during setup.

The second adapter was configured as a VirtualBox Host-only Adapter. This adapter was used for the internal lab network and was assigned the static IP address 192.168.56.10.

IP address: 192.168.56.10
Subnet mask: 255.255.255.0
Default gateway: blank
Preferred DNS server: 127.0.0.1

The command ipconfig was used to verify the final network configuration. The output confirmed that the NAT adapter and host-only adapter were both present, and that the host-only adapter had the correct static IP address.

```
C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : powerhub
    IPv6 Address. . . . . : fd17:625c:f037:2:a8e6:6f39:6736:7efb
    Link-local IPv6 Address . . . . . : fe80::a8e6:6f39:6736:7efb%6
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%6
                                10.0.2.2

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::55f7:beb2:7e8:39e%10
    IPv4 Address. . . . . : 192.168.56.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

Findings

The Windows Server virtual machine was installed successfully and configured for use in an Active Directory lab. The NAT adapter provided basic connectivity, while the host-only adapter created an isolated lab network for communication with the Windows client machine.

Assigning a static IP address to the server was important because domain controllers and DNS servers should use stable IP addresses. This allows client machines to reliably locate the domain controller when joining the domain and authenticating users.

Conclusion

This part of the lab successfully prepared the Windows Server virtual machine for Active Directory configuration. The server was installed, basic network settings were reviewed, and a static host-only IP address was configured for the lab environment. This provided the foundation for installing Active Directory Domain Services and creating the adlab.local domain in the next stage.